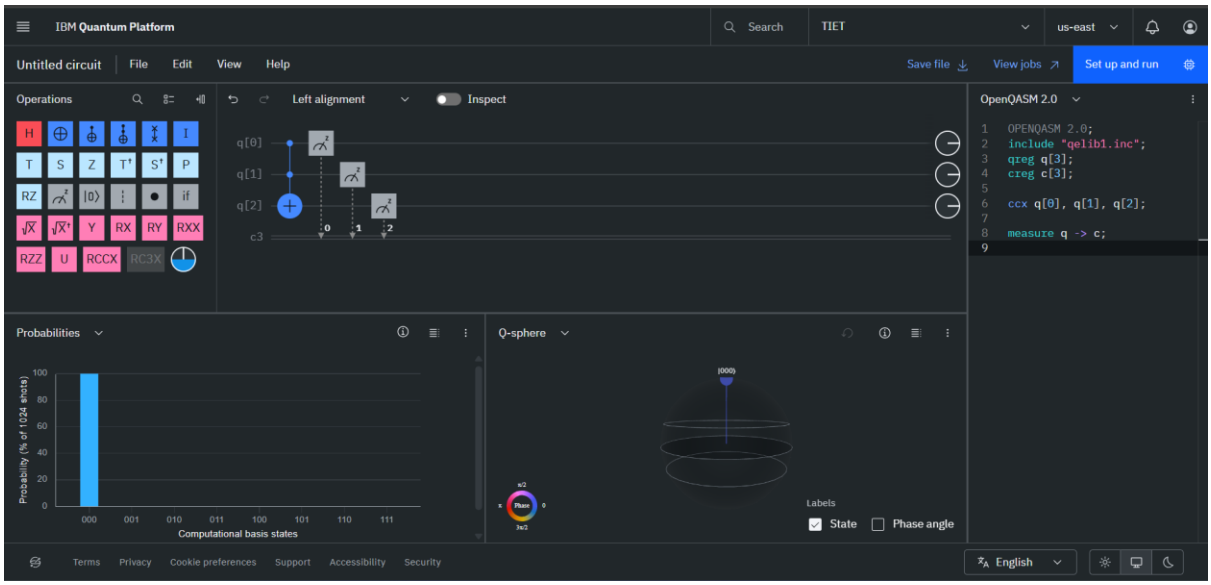
Assignment- 5

Roll no.:- 102316009

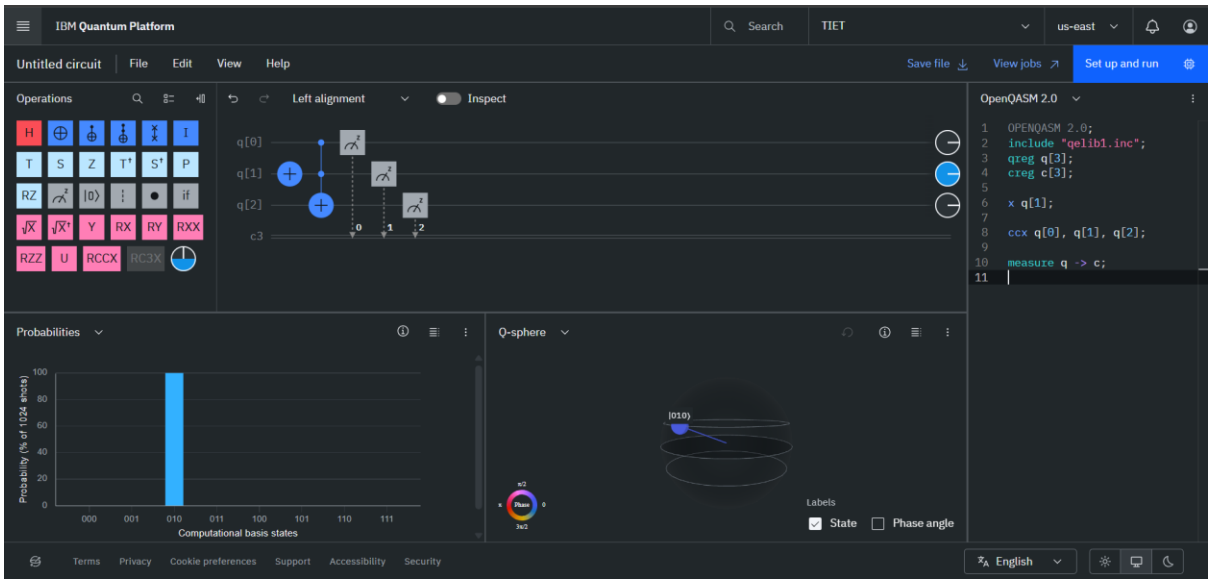
Experiment:- 1

AND Gate

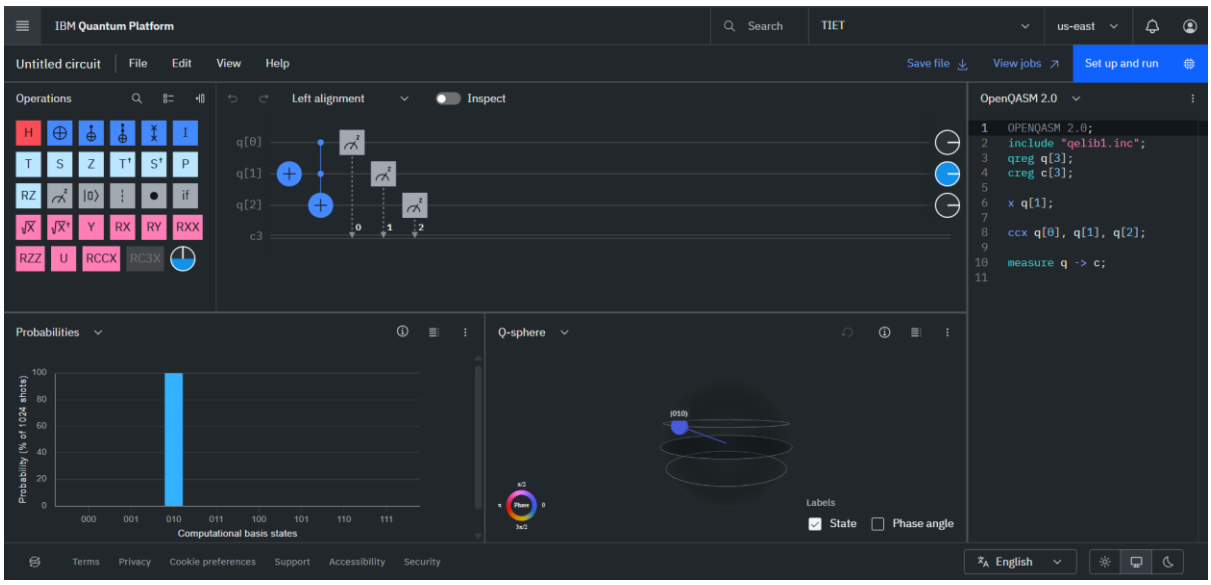
1. Input:- 00



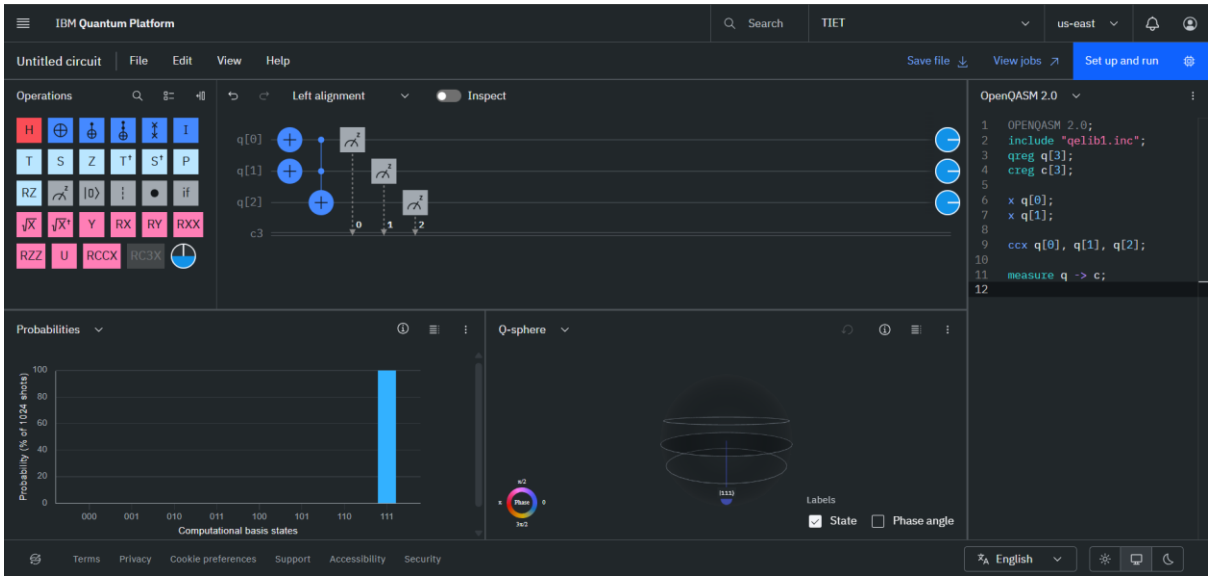
2. Input:- 01



3. Input:- 10

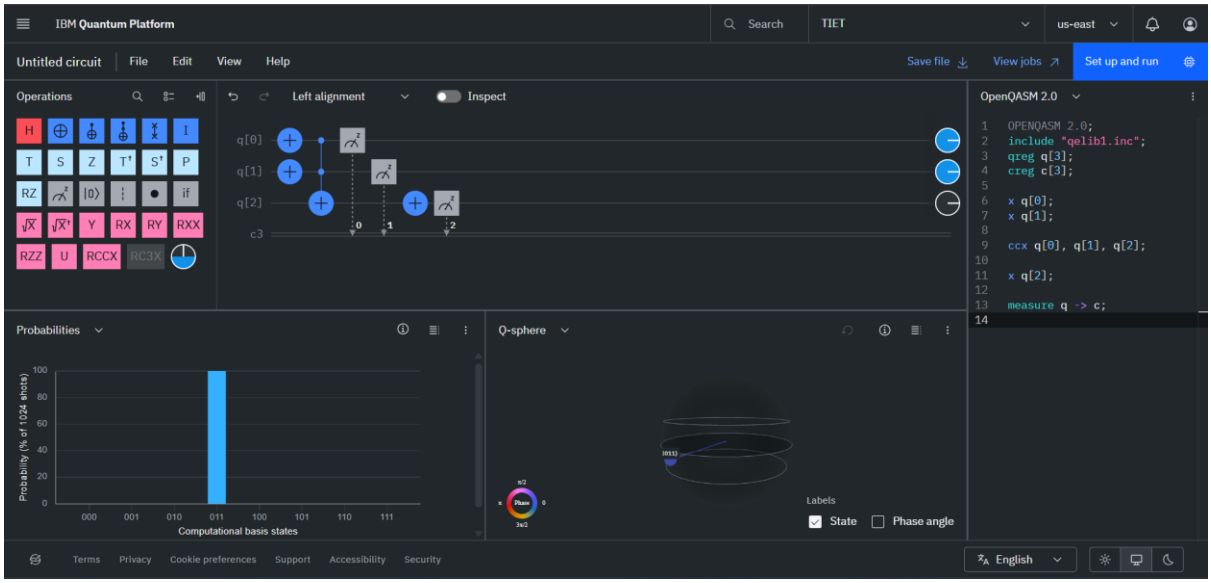


4. Input:- 11

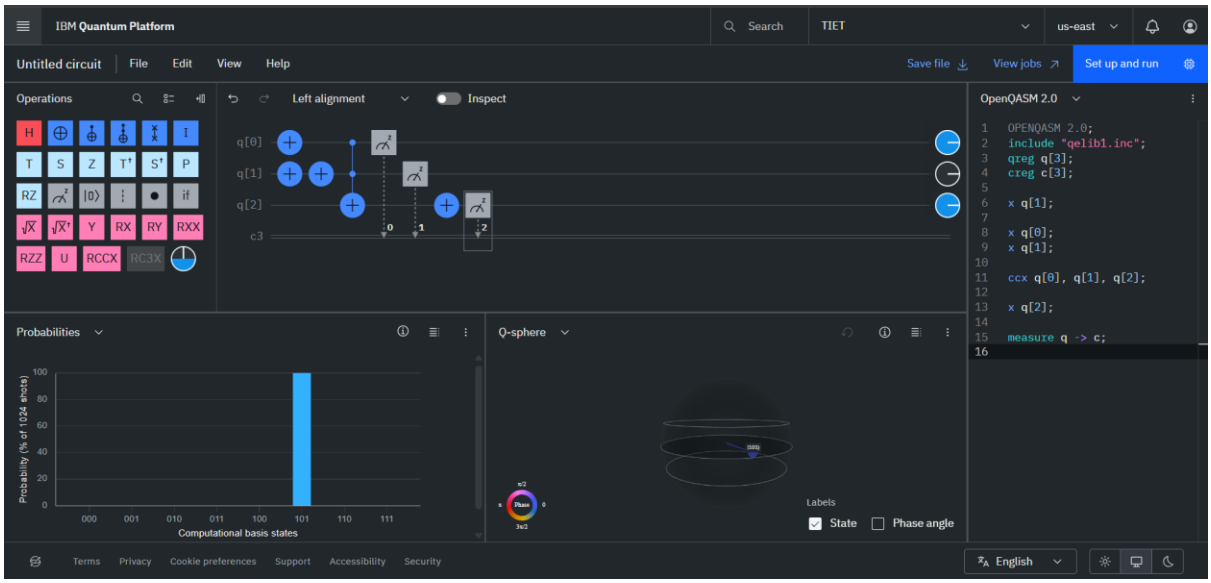


OR Gate

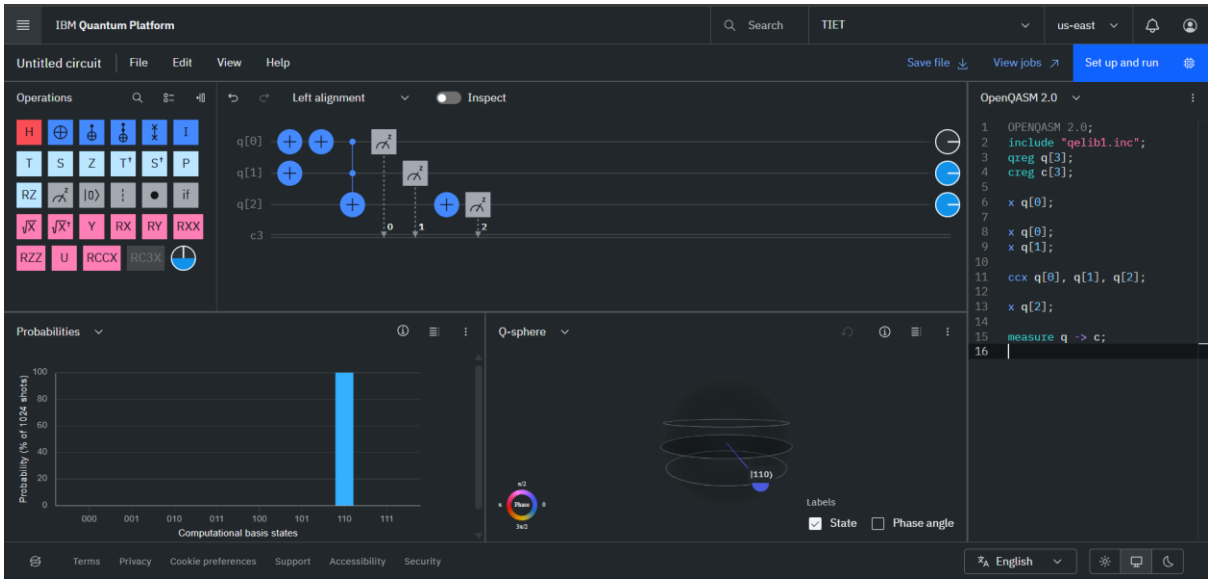
1. Input:- 00


IBM Quantum Platform
Search
TIET
us-east
Untitled circuit
File
Edit
View
Help
Save file
View jobs
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[3];
creg c[3];
x q[0];
x q[1];
ccx q[0], q[1], q[2];
x q[2];
measure q -> c;
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
Labels
State
Phase angle
Terms
Privacy
Cookie preferences
Support
Accessibility
Security
English

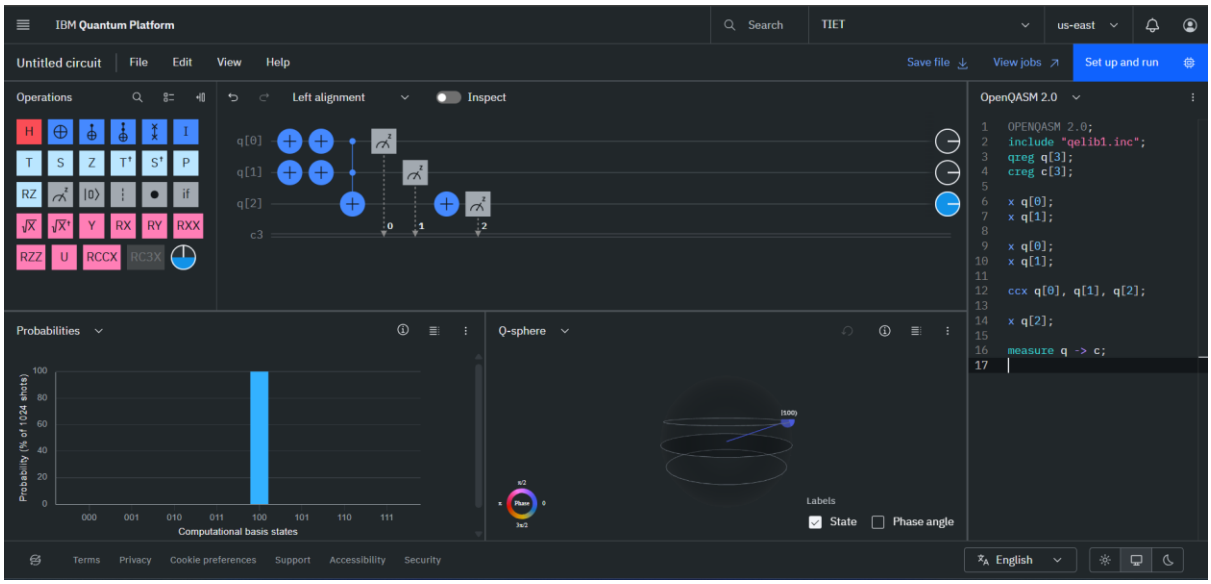
2. Input:- 01


IBM Quantum Platform
Search
TIET
us-east
Untitled circuit
File
Edit
View
Help
Save file
View jobs
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[3];
creg c[3];
x q[1];
x q[0];
x q[1];
ccx q[0], q[1], q[2];
x q[2];
measure q -> c;
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
Labels
State
Phase angle
Terms
Privacy
Cookie preferences
Support
Accessibility
Security
English

3. Input:- 10

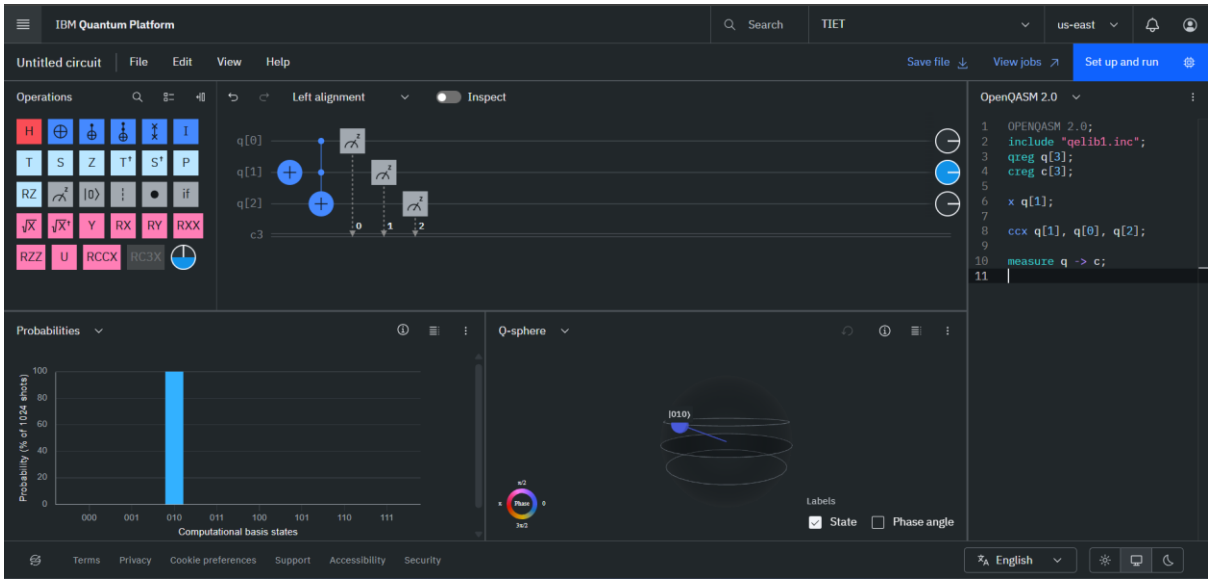


4. Input:- 11

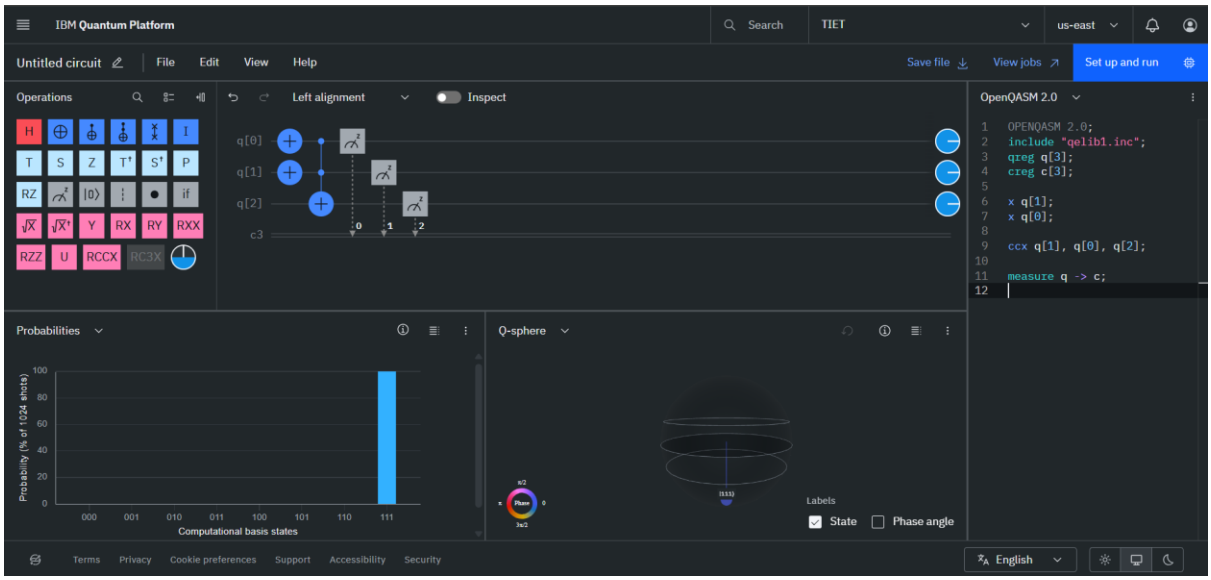


NOT Gate

1. Input:- 0



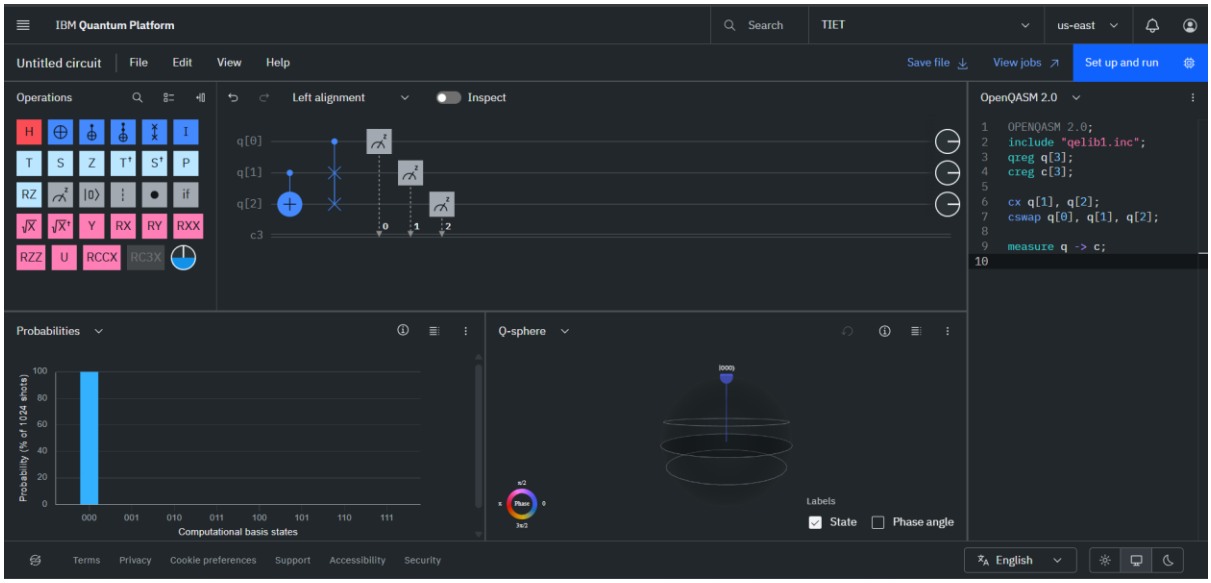
2. Input:- 1



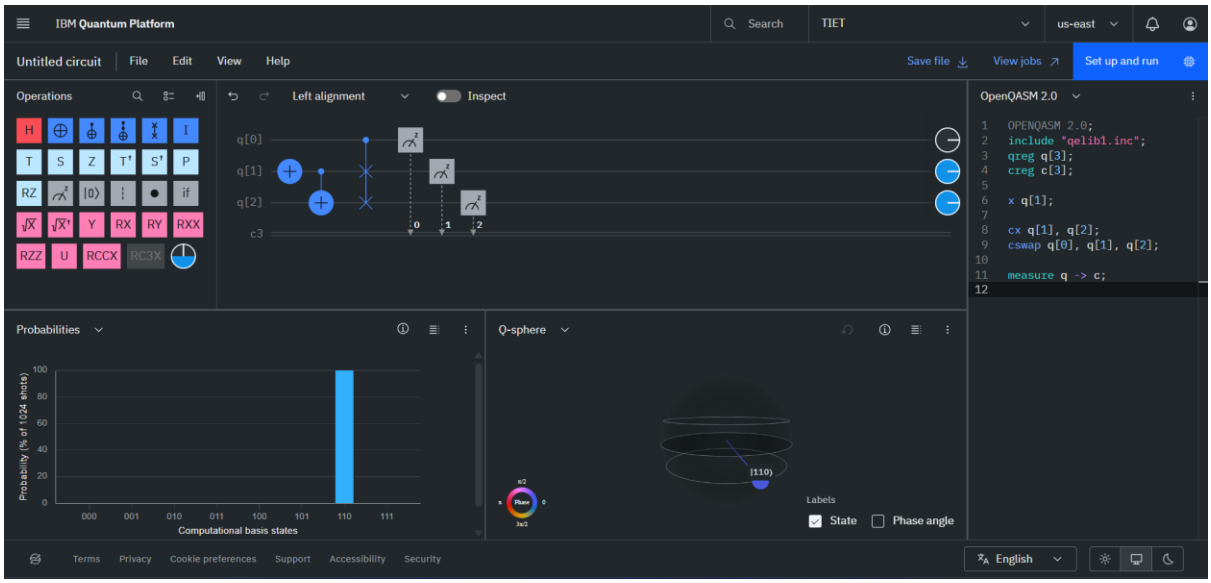
Experiment:- 2

AND Gate

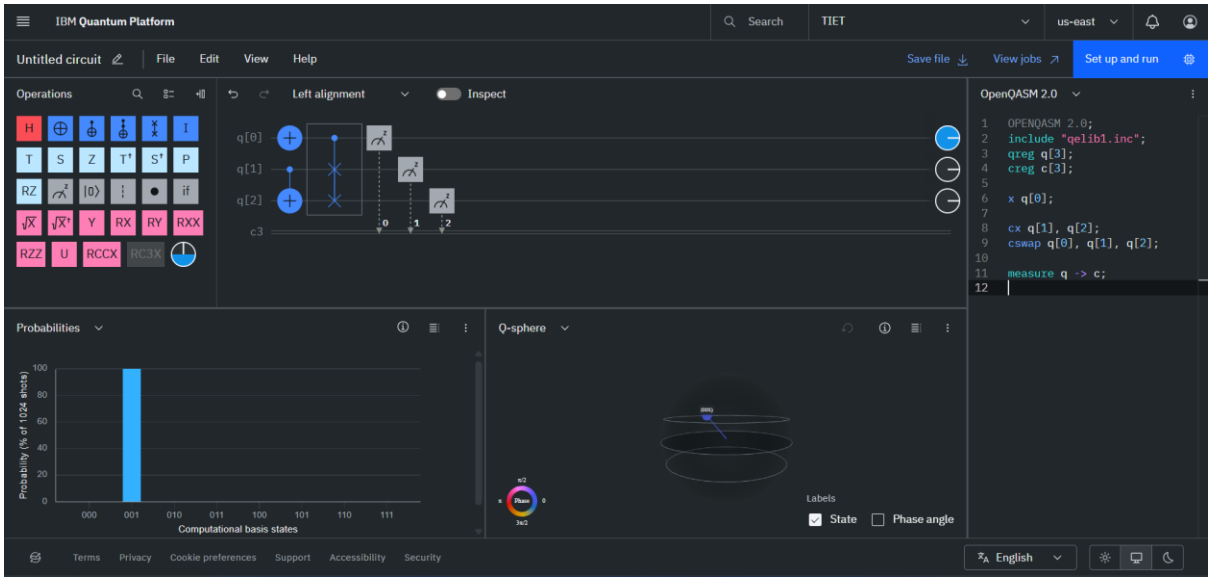
1. Input:- 00


IBM Quantum Platform
Search
TIET
us-east
Untitled circuit
File
Edit
View
Help
Save file
View jobs
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[3];
creg c[3];
cx q[1], q[2];
cswap q[0], q[1], q[2];
measure q -> c;
Probabilities
Q-sphere
Computational basis states
Labels
State
Phase angle
Terms
Privacy
Cookie preferences
Support
Accessibility
Security
English

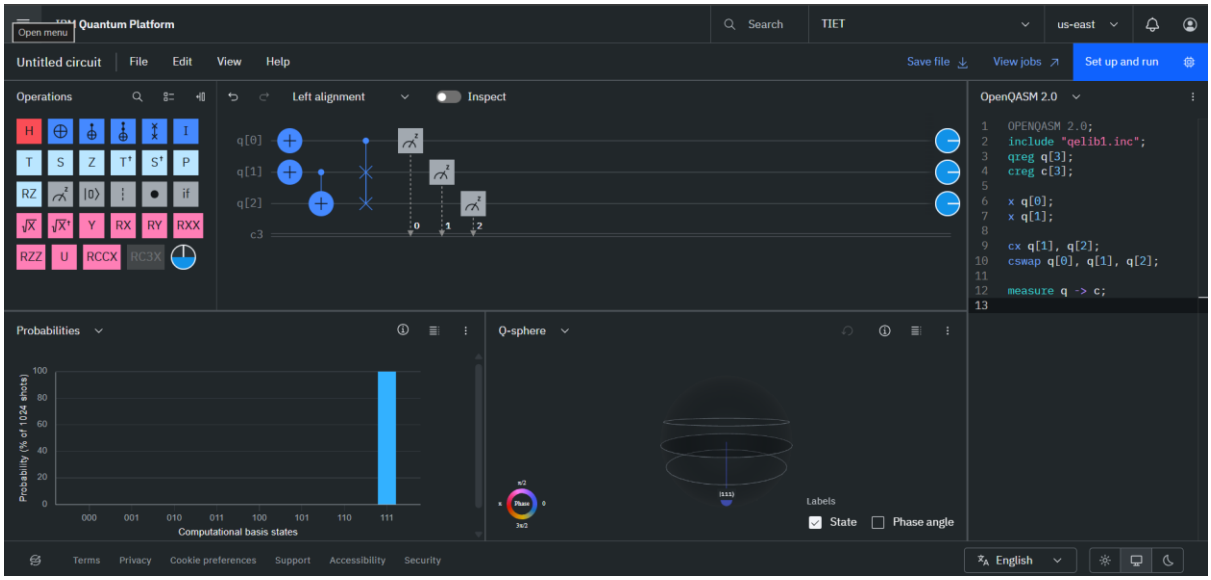
2. Input:-01


IBM Quantum Platform
Search
TIET
us-east
Untitled circuit
File
Edit
View
Help
Save file
View jobs
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
OPENQASM 2.0;
include "qelib1.inc";
qreg q[3];
creg c[3];
x q[1];
cx q[1], q[2];
cswap q[0], q[1], q[2];
measure q -> c;
Probabilities
Q-sphere
Computational basis states
Labels
State
Phase angle
Terms
Privacy
Cookie preferences
Support
Accessibility
Security
English

3. Input:- 10

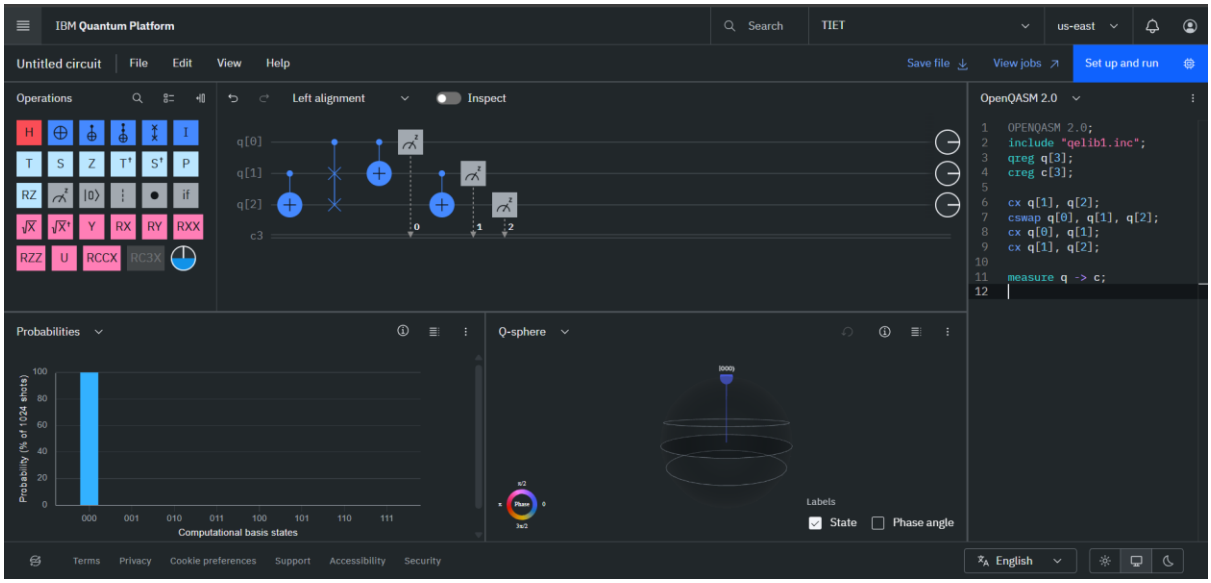


4. Input:- 11

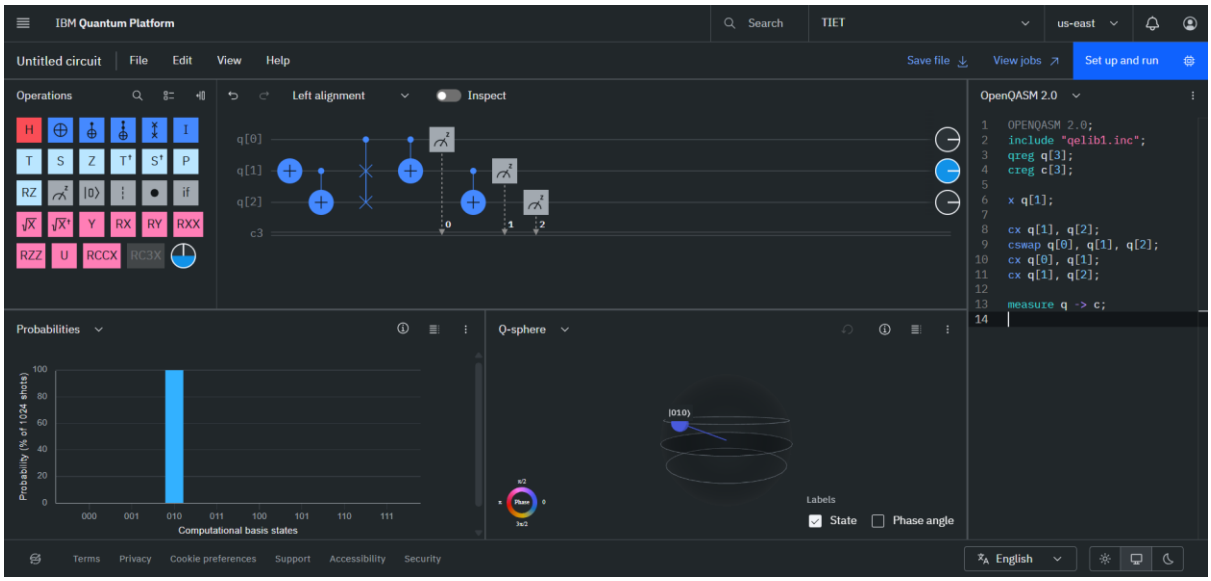


OR Gate

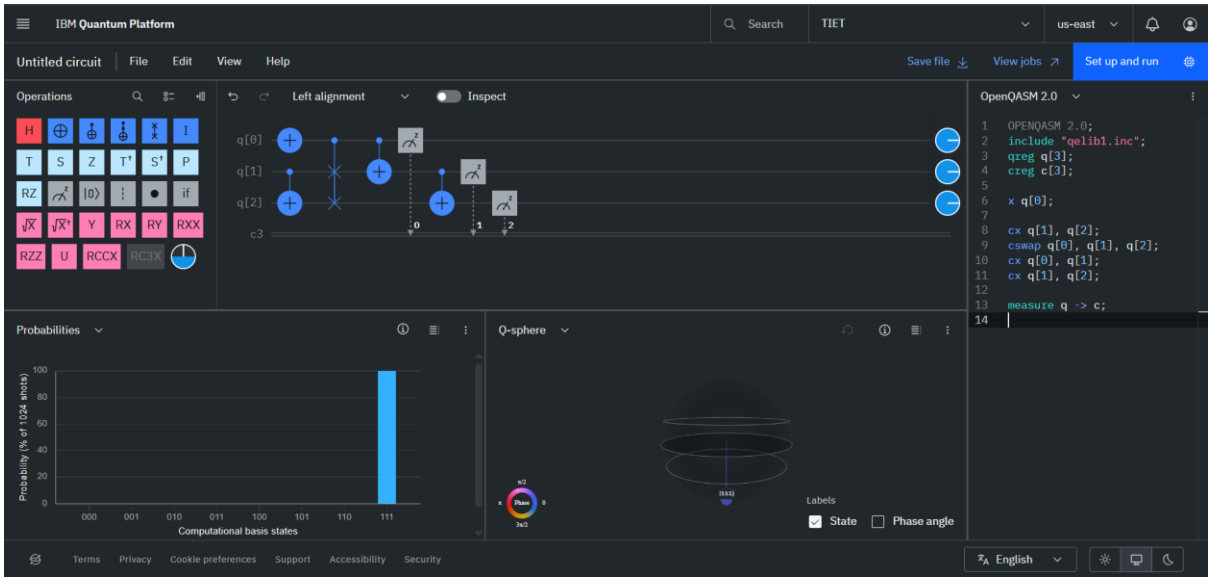
1. Input:- 00



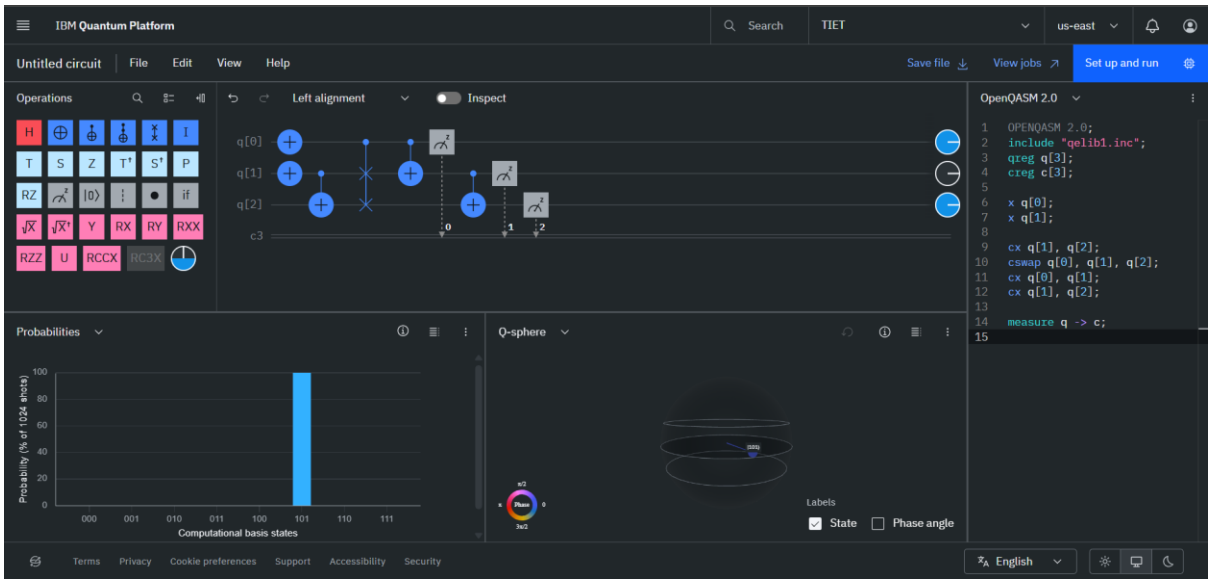
2. Input:- 01



3. Input:-10

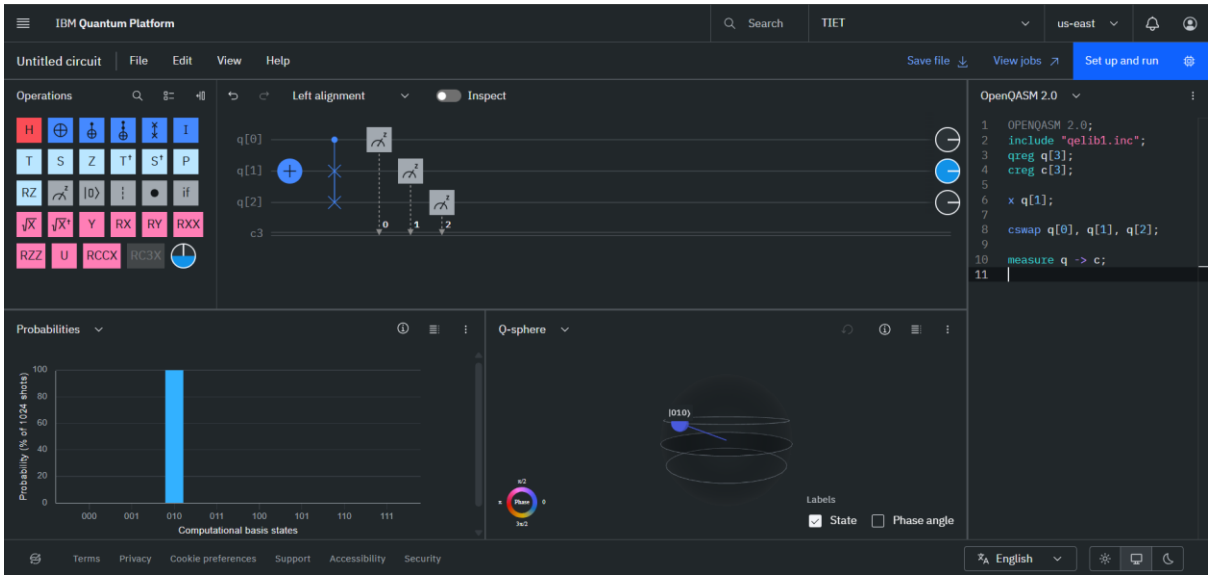


4. Input:- 11

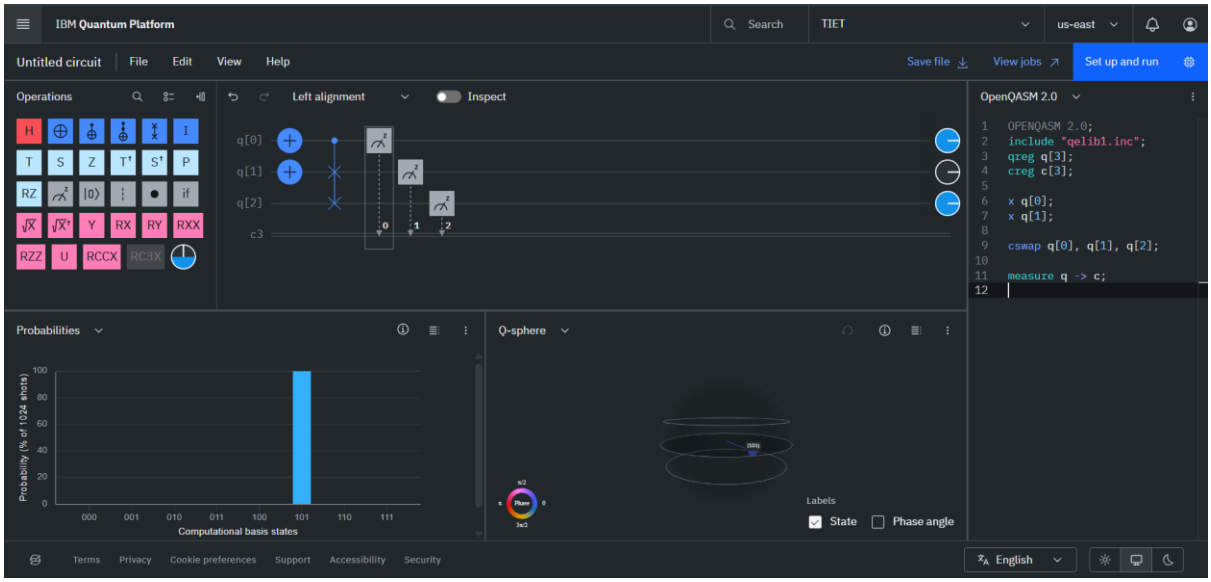


NOT Gate

1. Input:- 0



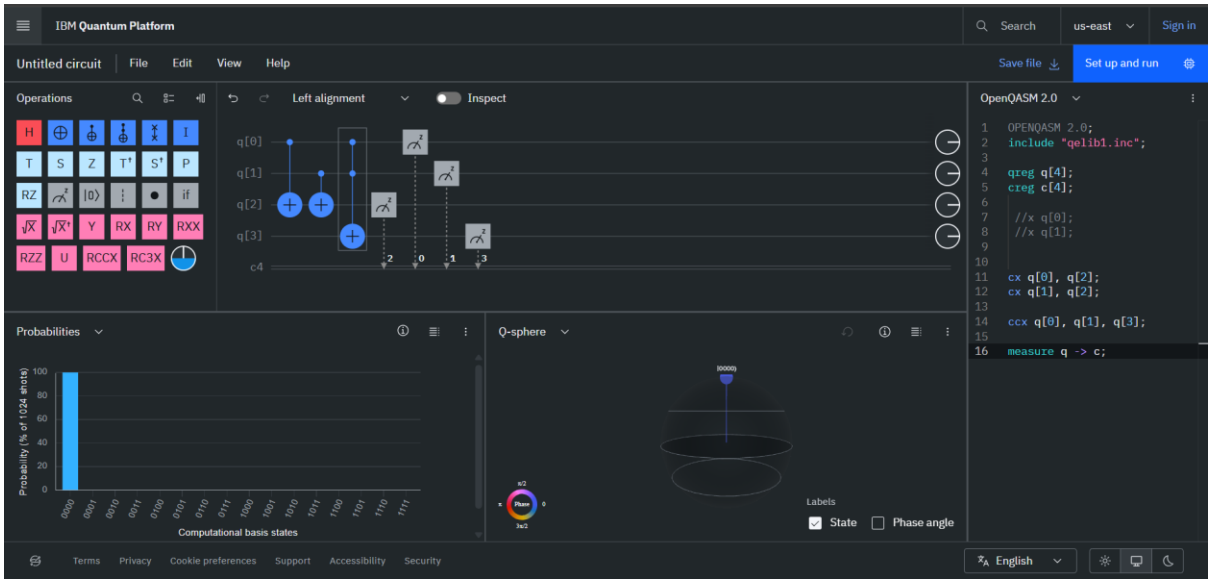
2. Input:- 1



Experiment 3

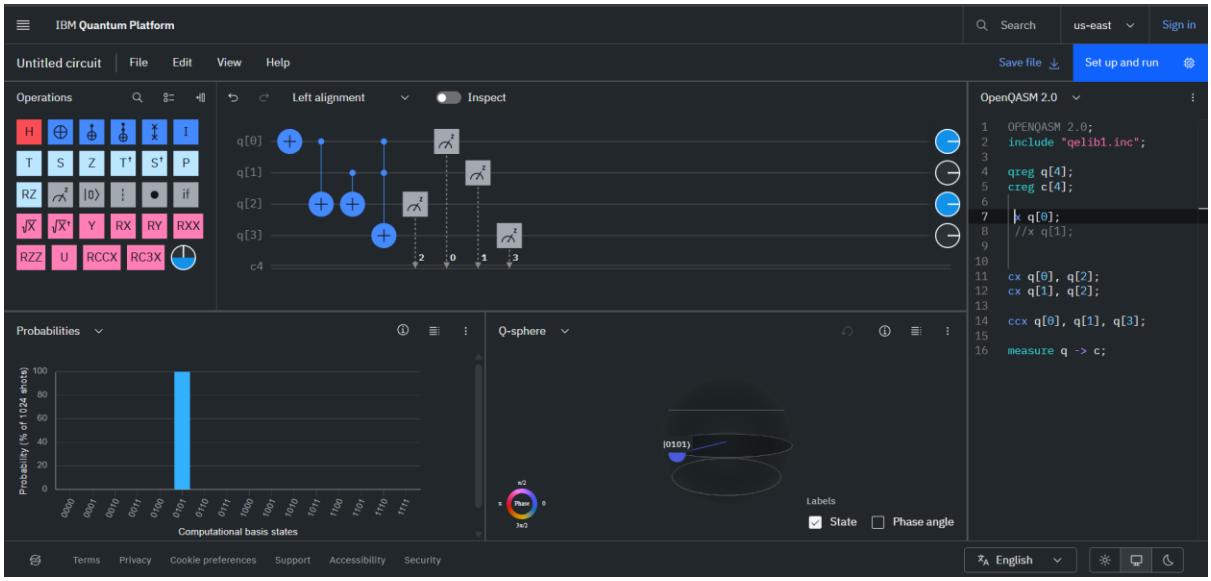
Half Adder

1. Input:- 00

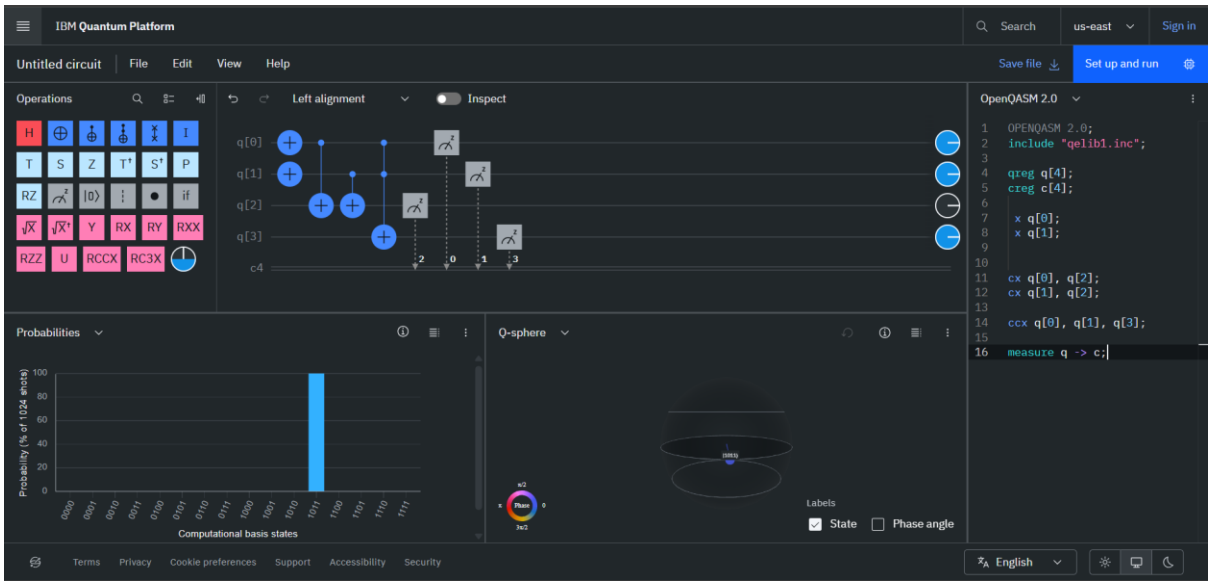


2. Input:- 01

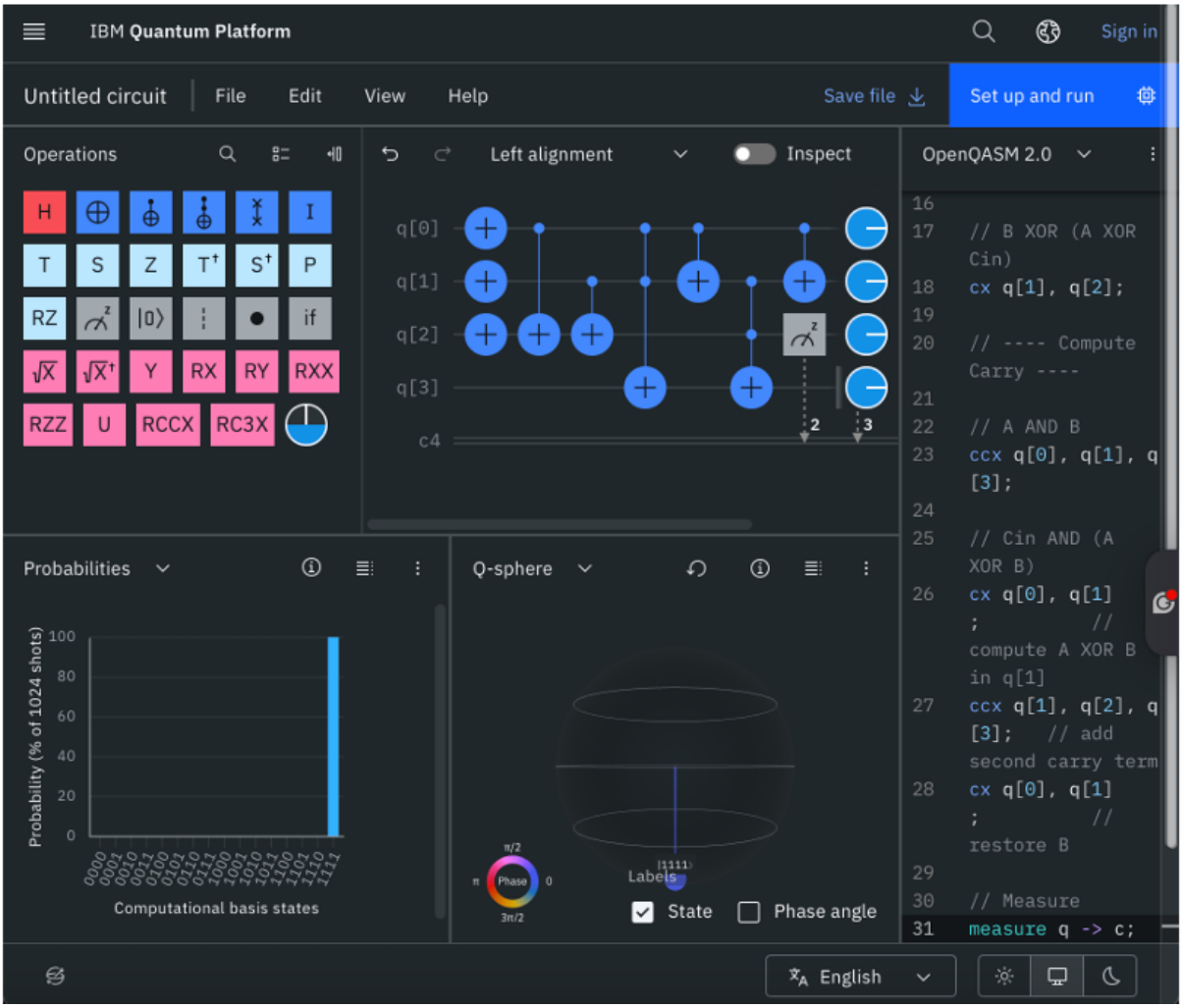
3. Input:- 10



4. Input:- 11

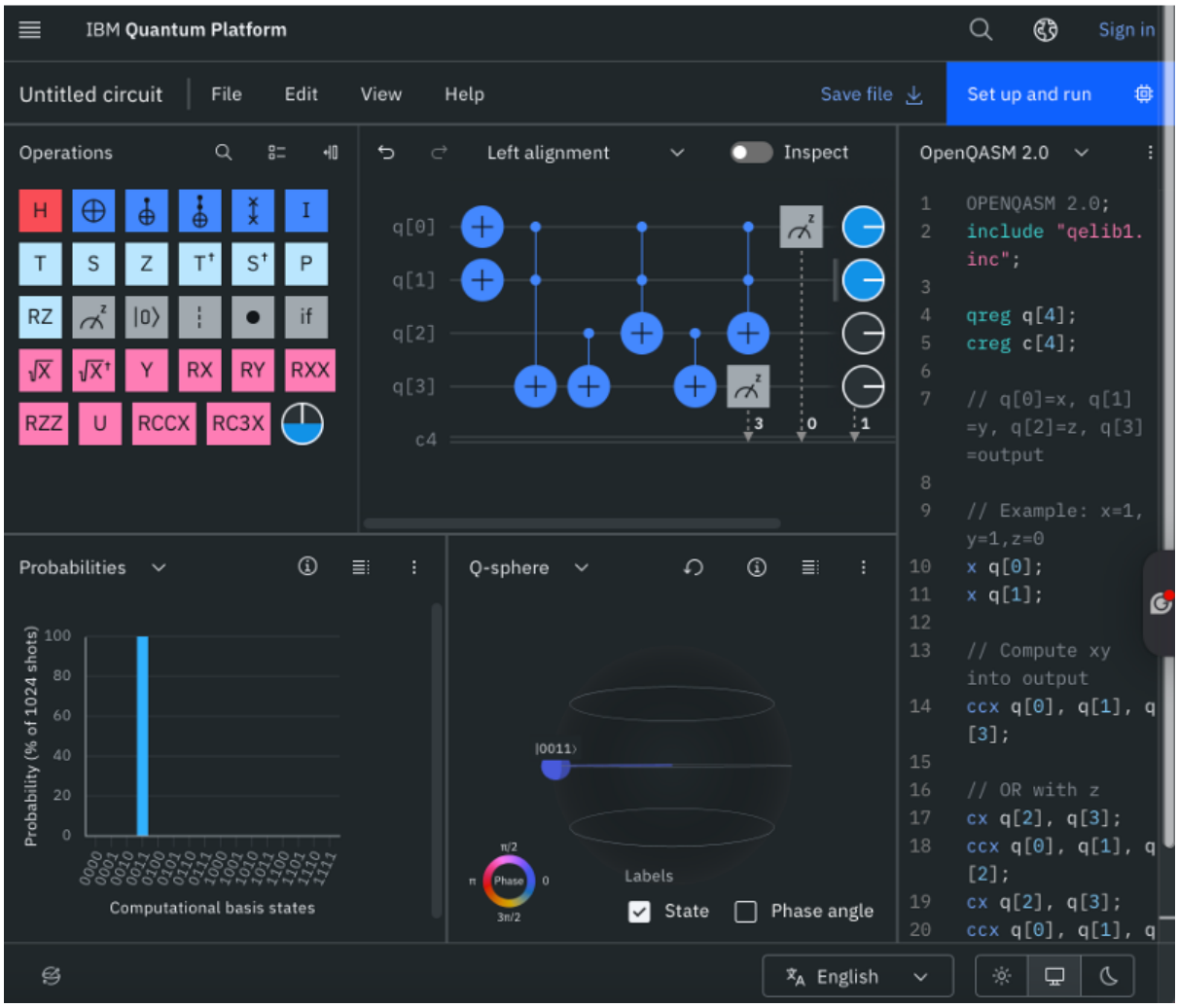


Full Adder

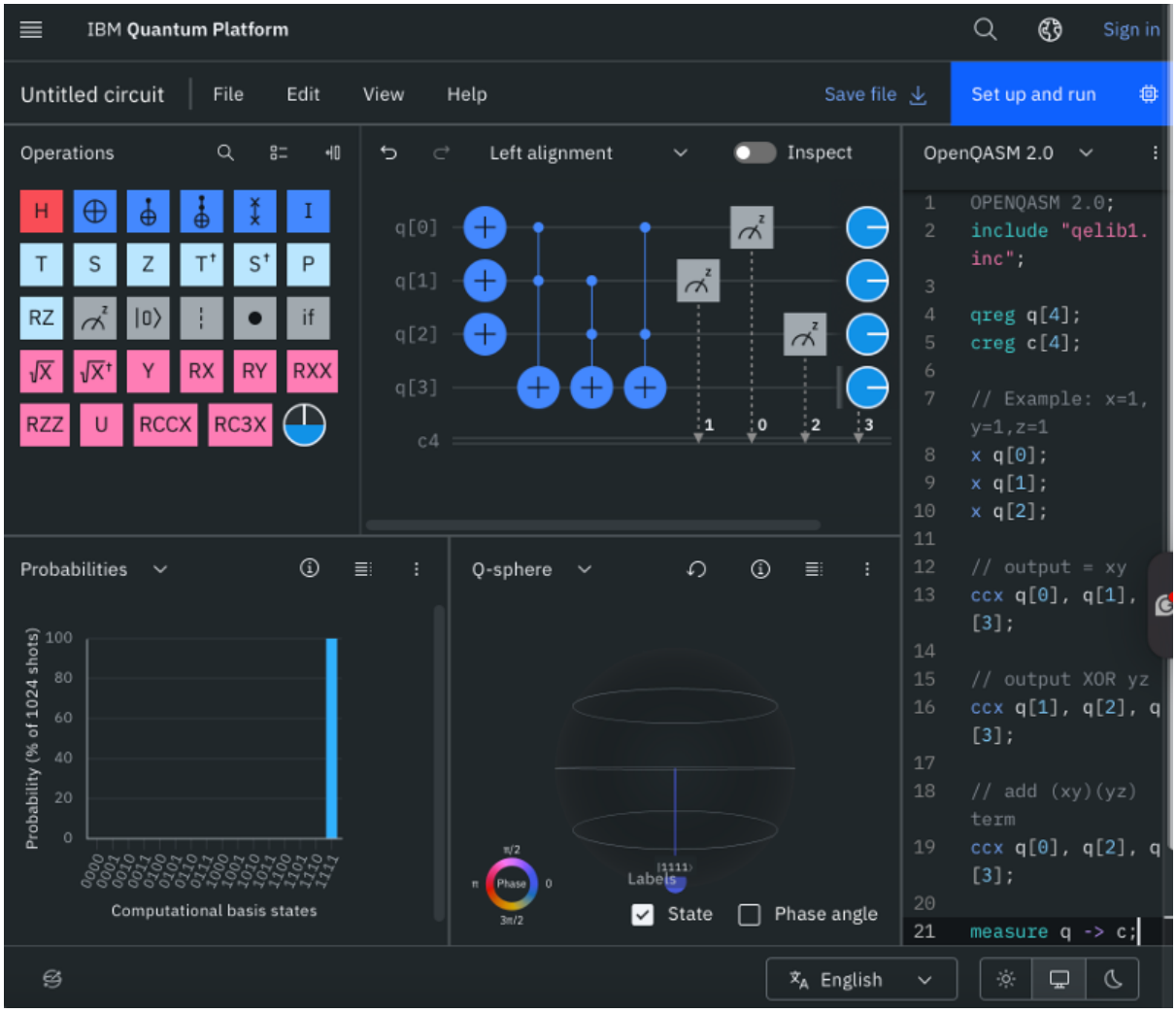


Experiment 4

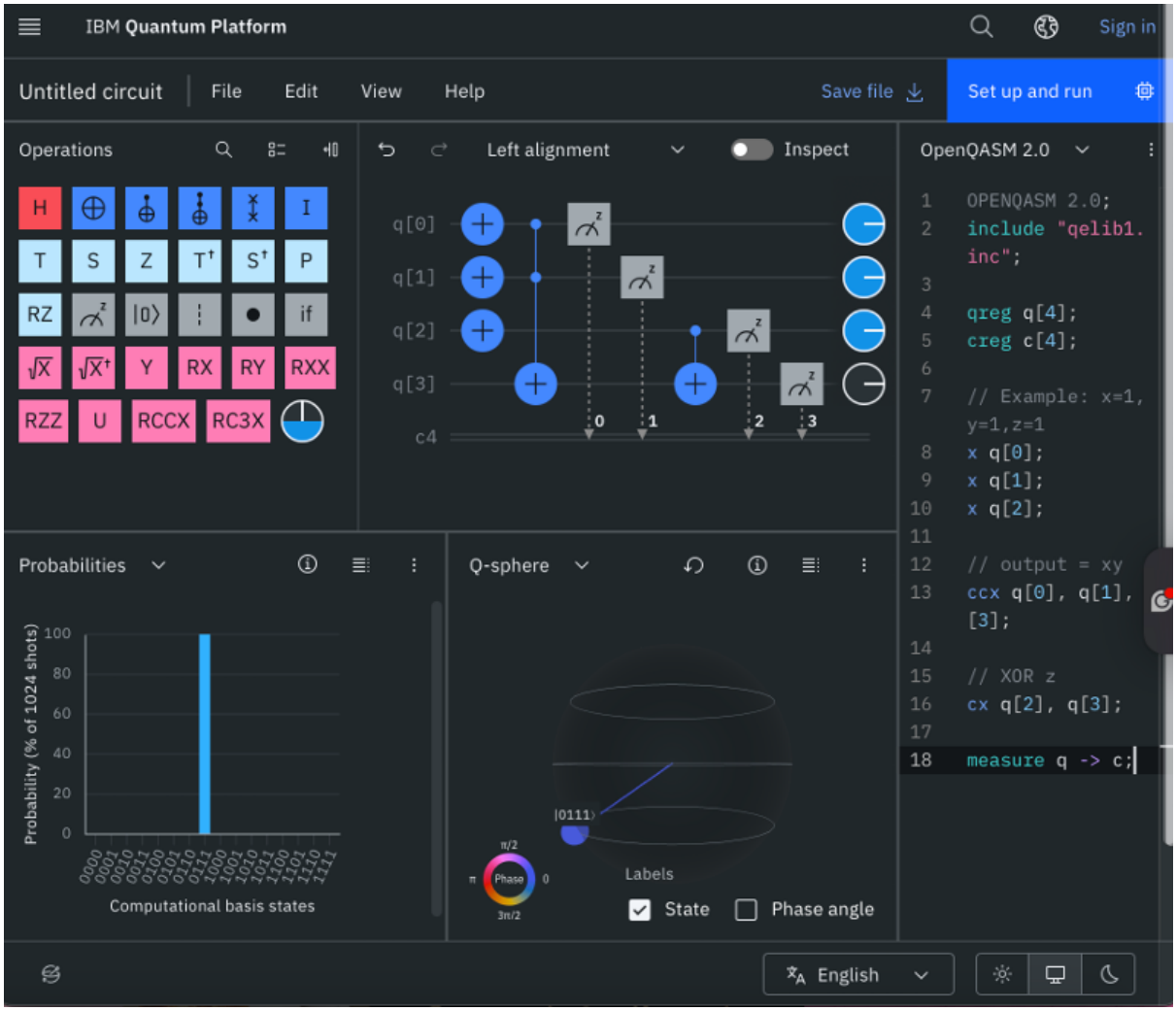
1. f(x,y,z) = (x ∧y)∨z


IBM Quantum Platform
Sign in
Untitled circuit
File
Edit
View
Help
Save file
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
q[0]
q[1]
q[2]
q[3]
c4
OPENQASM 2.0;
include "qelib1.inc";
qreg q[4];
creg c[4];
// q[0]=x, q[1]=y, q[2]=z, q[3]=output
// Example: x=1, y=1,z=0
x q[0];
x q[1];
// Compute xy into output
ccx q[0], q[1], q[3];
// OR with z
cx q[2], q[3];
ccx q[0], q[1], q[2];
cx q[2], q[3];
ccx q[0], q[1], q
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
|0011⟩
Labels
State
Phase angle
English

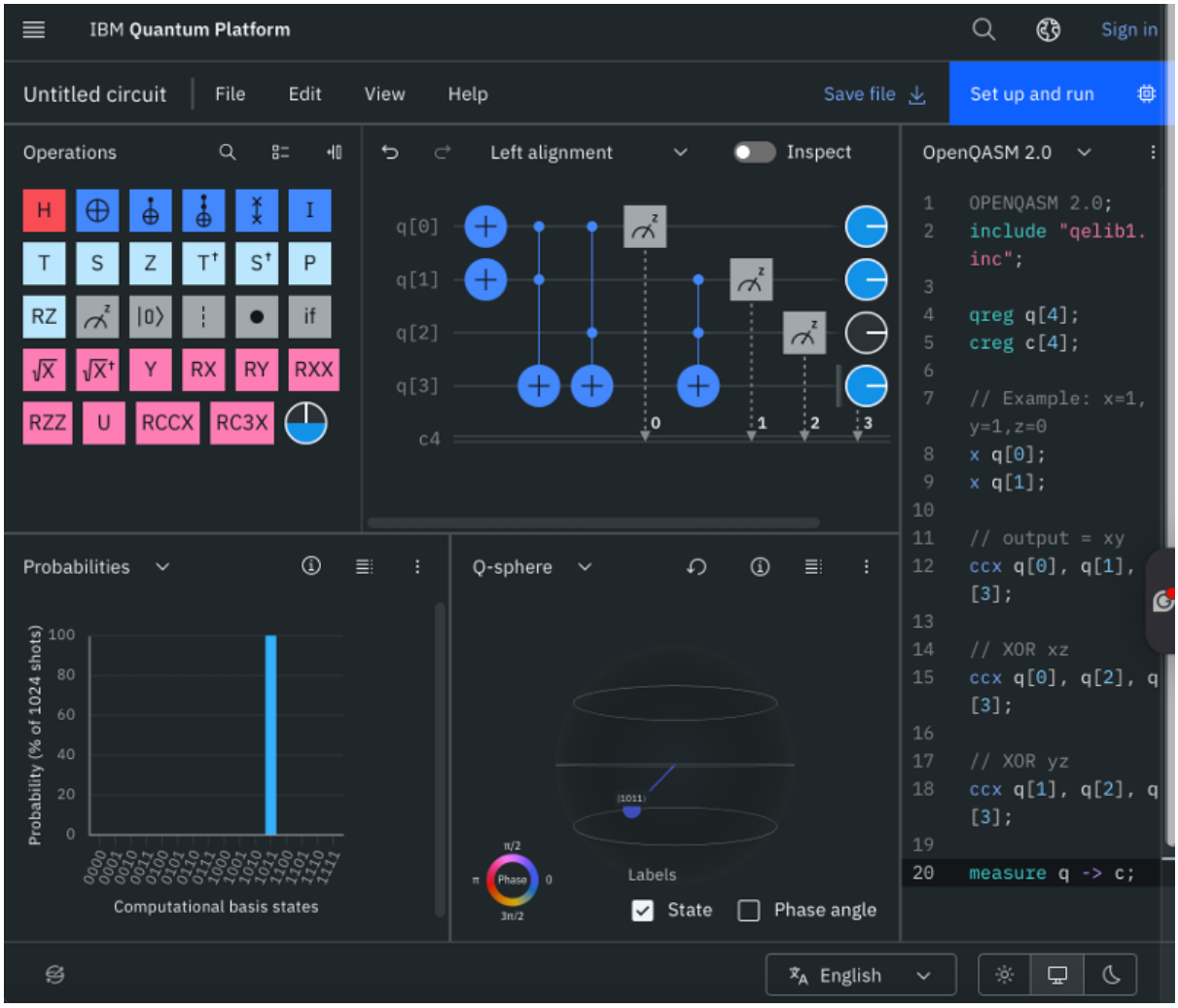
2. f(x,y,z) = (x ∧y)∨(y ∧z)


IBM Quantum Platform
Sign in
Untitled circuit
File
Edit
View
Help
Save file
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
q[0]
q[1]
q[2]
q[3]
c4
OPENQASM 2.0;
include "qelib1.inc";
qreg q[4];
creg c[4];
// Example: x=1, y=1,z=1
x q[0];
x q[1];
x q[2];
// output = xy
ccx q[0], q[1], [3];
// output XOR yz
ccx q[1], q[2], q[3];
// add (xy)(yz) term
ccx q[0], q[2], q[3];
measure q -> c;
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
Labels
State
Phase angle
English

3. f(x,y,z) = (x ∧y)⊕z



4. f(x,y,z) = (x ∧y)∨(x∧z)∨(y ∧z)


IBM Quantum Platform
Sign in
Untitled circuit
File
Edit
View
Help
Save file
Set up and run
Operations
Left alignment
Inspect
OpenQASM 2.0
q[0]
q[1]
q[2]
q[3]
c4
OPENQASM 2.0;
include "qelib1.inc";
qreg q[4];
creg c[4];
// Example: x=1, y=1,z=0
x q[0];
x q[1];
// output = xy
ccx q[0], q[1], [3];
// XOR xz
ccx q[0], q[2], q[3];
// XOR yz
ccx q[1], q[2], q[3];
measure q -> c;
Probabilities
Probability (% of 1024 shots)
Computational basis states
Q-sphere
Labels
State
Phase angle
English